Hom Degree Forms: deterministic pilot observations

Observation-only status

This report records execution outputs; it makes no support, weakening, priority or scientific-status decision. Independent mathematical review remains separate. All instances are supplied synthetic lattices.

Trial and provenance

Experiment EXP-ENDO-371597; hypothesis H-ENDO-d492e4; run RUN-ENDO-859fbd. Code commit 443652caa4102d2007aa48418649f7b239924ee2. Seed 1729. Exact command: ["python3", "tools/lattice_hom_degree.py", "--seed", "1729", "--output-dir", "/workspace/lattice-hom/experiments/EXP-ENDO-371597/runs/RUN-ENDO-859fbd"]. One trial requested; completion/failure and resource observations are preserved in execution-receipt.json. The driver records full-tree dirtiness as unknown when not supplied; every source byte is admitted against its committed hash.

Recorded outputs

Exact fixture counters (integer counts, not independent statistical samples): {"all_exact_checks_passed": true, "form_cases": 5760, "pair_cases": 1152, "triangle_cases": 220}. Supervisor receipt: {"artifact_sha256": {"check.stderr.log": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855", "check.stdout.log": "81231d85125512240ad58415f71c858fcde28f61b4f58a131ae2f4141ec842f0", "command.txt": "d455db804e228f7302eff3e70ae477715dc2af87a64a2016906020de92c2c62c", "environment.json": "cf5734aec980c12287da7f66c7f5bbb65e7c4c207284fc0d58b1981759294ade", "launch.json": "9def1595cc840b8ee12e60fa58ecb1cd2000842fa8d71aa5d9dff7430fd20c3d", "manifest.yaml": "6e08acbc276dc1dd99e316b8ebeba503f9393967f420d346147de85db887a70d", "raw-result.json": "72fb5c0cc2f3e4089b7169182b645ad6773bb6cab6c4dca9a413676cbe20194f", "stderr.log": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855", "stdout.log": "ed7e535333f0afaa6e8a8149300d7da776695856bad60d21607ea632925da12a"}, "authority": {"commit": "443652caa4102d2007aa48418649f7b239924ee2", "dispatch_plan_sha256": "9818f8421453ada490d3024ec9cdcd4357a18ebbfe98156f119883bd9342ee9e", "epoch": 1, "expires_at": "2026-10-14T16:13:15+00:00", "owner": "lattice-executor-20261014"}, "check_returncode": 0, "detail": null, "finished_at": "2026-10-14T15:16:39.158581+00:00", "plan_sha256": "363376c017006fb91cf4b2e8c92a93c6248ce9eca494a59eebc8f1af29ec2046", "published": false, "returncode": 0, "run_id": "RUN-ENDO-859fbd", "schema": "crypto.autoresearch.trial_receipt.v1", "scientific_conclusion": null, "started_at": "2026-10-14T15:16:38.744457+00:00", "status": "output_validated", "trial_id": "exact-pilot-seed1729"}

Control scope and graph

One retained triangle (6,10,12); scaled product-only answer is rejected by the full-form gate. All plotted values come directly from raw-result.json; no new sampling is performed. Uncertainty bars are inapplicable to these deterministic exact values. Plot source and extracted data are retained.

Limitations and missing data

Only the principal supplied order lattices C/O_f are implemented. Full Hom bases and their control sublattice indices are planted and certified algebraically. Triangle consistency cannot certify completeness of an unknown Hom lattice. No finite-field maps, map-discovery cost, performance advantage, or ECDLP consequence was measured. The integrity checker replays the implementation; it is not independent research review. No ECDLP canonical graph or frontier was changed.

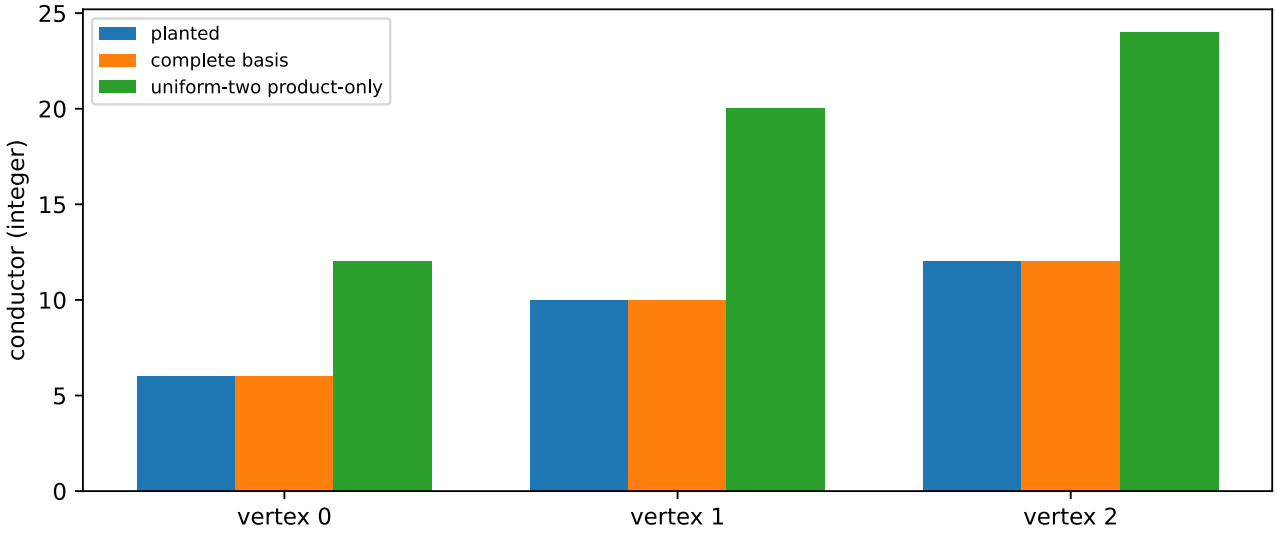

Figure 1: One retained triangle (6,10,12); scaled product-only answer is rejected by the full-form gate. Run RUN-ENDO-859fbd.